

Mahir Ahmed

Final_FYDP_Report (Team_Auditory_Cortex_)

 FYDP 123 Report

Document Details

Submission ID

trn:oid:::3618:142631955

Submission Date

Jun 11, 2026, 8:19 PM GMT+6

Download Date

Jun 11, 2026, 11:21 PM GMT+6

File Name

Final_FYDP_Report (Team_Auditory_Cortex_).pdf

File Size

1008.7 KB

44 Pages

10,055 Words

60,703 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Onubad: An AI-Based Bangla Sign Language Recognition System and Translation into Text & Speech

Submitted By:

Mahir Ahmed	0112230768
Tahmid Rahman Osmani	0112230733
Md. Azmain Sakin	0112230734
Mahfuzul Islam Pranto	0112230767
Md. Rayhan Islam Showrav	0112230810

Submitted in partial fulfilment of the requirements
of the degree of Bachelor of Science in Computer Science and Engineering

June 11, 2026



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNITED INTERNATIONAL UNIVERSITY

Abstract

There exist a massive number of deaf and people with hearing disabilities in Bangladesh who have a communication gap with other. Moreover it lack technological, realtime support for Bangla Sign Language (BdSL), as most of the existing solutions are word level and lacks context awareness while sentence translation. Our project, titled "Onubad" will try to addresses this gap by simply making a translator hence the name Onubad. Our intelligent system will translates BdSL gestures into meaningful Bangla text. Our proposed methodology will try to utilize a Dual Stream Semantic Fusion Architecture[1], which uses MediaPipe[2] to extract gesture landmarks from realtime video inputs from the users then the system uses a parallel stream procedure by using a Transformer based[3] word spotter for frequently used word signs and a YOLO Nano [4]based character spotter for fingerspelling characters. Then both these output streams are merged and processed by a quantized edge level Large Language Model (LLM) [5]that performs semantic reconstruction which will ensure grammatical accuracy and make contextually meaningful sentence.

Acknowledgements

Publication List

[Since the research is still on progress, thus no publications yet.]

Journal Articles

- 1.

Conference Papers

- 1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

- 1.

Table of Contents

Table of Contents	v
List of Figures	vi
List of Tables	vii
1 Introduction	1
1.1 Project Overview	1
1.2 Motivation	2
1.3 Objectives	2
1.4 Methodology	3
1.4.1 Feature Extraction	3
1.4.2 Dual-Stream Parallel Recognition	3
1.4.3 LLM Semantic Reconstruction	4
1.5 Project Outcome	4
1.6 Organization of the Report	4
2 Background	5
2.1 Preliminaries	5
2.2 Literature Review	6
2.2.1 Similar Applications	6
2.2.2 Related Research	6
2.2.3 Summary of Literature	9
2.3 Gap Analysis	15
2.4 Summary	15
3 Project Design	16
3.1 Requirement Analysis	16
3.1.1 Functional and Nonfunctional Requirements	16
3.1.2 Context Diagram	17
3.1.3 Data Flow Diagram Level 1	17
3.1.4 UI Design	19
3.2 Detailed Methodology and Design	20

Table of Contents

Table of Contents

3.3	Project Plan	21
3.4	Task Allocation	21
3.5	Summary	22
4	Implementation and Results	23
4.1	Environment Setup	23
4.2	Testing and Evaluation	23
4.3	Results and Discussion	23
4.4	Summary	23
5	Standards and Design Constraints	24
5.1	Compliance with the Standards	24
5.1.1	Software Standards	24
5.1.2	Hardware Standards	24
5.1.3	Communication Standards	25
5.2	Design Constraints	25
5.2.1	Economic Constraint	25
5.2.2	Environmental Constraint	25
5.2.3	Ethical Constraint	25
5.2.4	Health and Safety Constraint	26
5.2.5	Social Constraint	26
5.2.6	Political Constraint	26
5.2.7	Sustainability	26
5.3	Cost Analysis	26
5.3.1	Proposed Budget	26
5.3.2	Alternate Budget and Rationale	27
5.3.3	Revenue Model	27
5.4	Complex Engineering Problem	28
5.4.1	Complex Problem Solving	28
5.4.2	Engineering Activities	29
5.5	Summary	30
6	Conclusion	31
6.1	Summary	31
6.2	Limitation	31
6.3	Future Work	31
	References	36

List of Figures

1.1	Overall Workflow of the Proposed System	3
3.1	Context Diagram of the Onubad Architecture	17
3.2	Level-1 Data Flow Diagram (DFD) of the Project "Onubad"	18
3.3	Detailed Level-2 Data Flow Diagram (DFD) for the Project "Onubad" . . .	19
3.4	Gantt chart of the Project Onubad	21

List of Tables

2.1	Summary of Related Research Papers	10
3.1	Design Rationale and Trade-off Evaluation Matrix	20
5.1	Proposed Budget for Onubad Project	27
5.2	Alternate Low-Cost Budget	27
5.3	Mapping with complex problem solving.	28
5.4	Mapping with complex engineering activities.	29

Chapter 1

Introduction

This chapter is an overview of our entire proposed project ,with the problem background ,motivation of the project and the main objectives. The proposed methodology, expected outcomes and overall structure of the report are also briefly described in the report.

1.1 Project Overview

In many parts of the world deaf and hearing individuals face a significant challenge to communicate,mainly in Bangladesh technological support and awareness for Bangla Sign Language are still limited.As sign language is a primary mean of communication for the deaf community and very few people know sign language outside the community,it creates a gap of communication necessary situations such as education,healthcare,daily life communication and public services.But existing systems for sign language detection are often limited to word level detection and alphabets.Thus,they do not provide a complete ,real time solution for natural sentence construction with semantic meaning.

This project, titled **“Onubad: An AI-Based Bangla Sign Language Recognition System”**, aims to solve this problem by developing a system which is capable of recognizing BdSL gestures and translating them into meaningful sentences and speech. The proposed system will take input from real time webcam feeds and also uploaded video clips, making it suitable for different use case scenarios. The output will include both textual translation of sentences and speech generation using text-to-speech (TTS), with the integration of a quantized language model to improve contextual accuracy,semantic meaning and next word suggestions. The final implementation is a web application or system built in devices, which will allow users to interact through a browser or directly with the device. This project aims to build an real time BdSL translation system which will also be accessible,scalable and can help reduce communication problems and promote and inclusive society.

1.2 Motivation

In Bangladesh, deaf and hard of hearing individuals commonly face communication problems significantly in classrooms, hospitals, and public services and daily conversations, as Bangla Sign Language (BdSL) is rarely understood by very few people. Although there is integration of artificial intelligence in sign language recognition, almost all research is focused on well resourced languages like American Sign Language. BdSL remains Overlooked, while existing solutions are limited to isolated signs rather than practical, sentence level translation and word predictions. This project solves that problem by applying multimodal approaches, computer vision and LLM reasoning to create a real time BdSL translation system with highly predictable sentence and speech output. The goal of this project is social and technical which is to reduce the communication barriers for an overlooked community and integration of Large Language Models with quantization for semantic and natural sentence building.

1.3 Objectives

The main objective of this project is to build an accessible, scalable, low latency Bangla Sign Language (BdSL) translation system which is designed to bridge the communication problems between deaf and hearing individuals. For achieving this goal, the project is centered around the given below objectives:

- **Efficient Spatial-Temporal Feature Extraction:** To extract privacy preserving and computationally efficient skeletal landmarks from video inputs using pose estimation frameworks such as MediaPipe. This approach lessens the impact of background noise and lighting variations while maintaining low hardware requirements.
- **High Accuracy Sign Detection:** Building a transformer based model which is capable of analyzing temporal dependencies in BdSL. This will ensure accurate detection of BdSL signs.
- **Multi modality fusion:** Merging additional gestures such as face expressions, movements of lips, and hand position. This is the multimodal approach to improve detection of complex and continuous sign language.
- **Sentence creation with context awareness:** Integrating a quantized LLM for creating context aware, grammatically correct sentences which will lead this project to simple word translation to real time context aware system.
- **Scalable and Accessible Deployment:** Building a user friendly web application or integrated system with UI in devices capable of running the architecture without any hassle.

1.4 Methodology

The proposed methodology of this project is mainly a dual stream semantic fusion architecture, which differs from traditional pattern matching architectures. Instead of depending only on feature extraction and model prediction, the system also adds a linguistic reasoning by large language model to produce more natural and meaningful outputs. The overall architecture is divided into three main sequential phases.

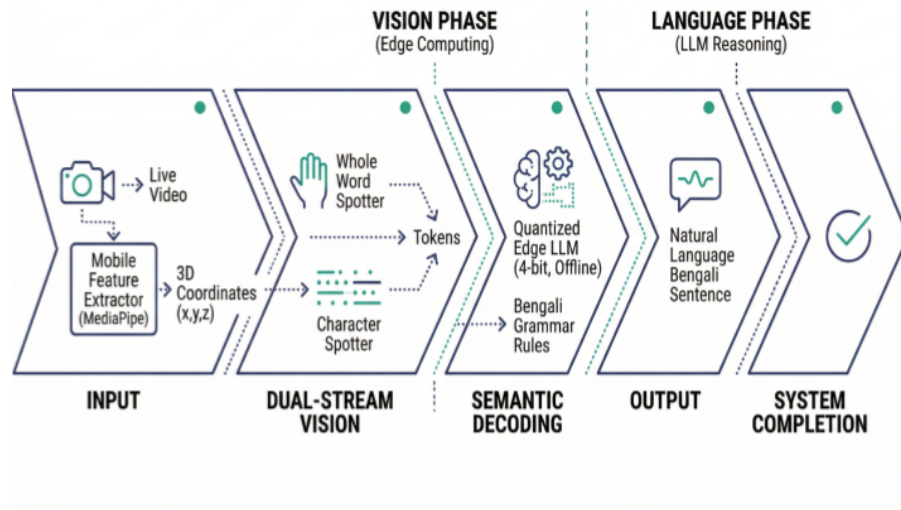


Figure 1.1: Overall Workflow of the Proposed System

1.4.1 Feature Extraction

Firstly, the system captures input from a real time video stream of 30 fps or from uploaded video. The captured frames are processed using MediaPipe to extract key skeletal landmarks of sign languages and gestures and converted into structured vector coordinates.

1.4.2 Dual-Stream Parallel Recognition

After feature extraction, the coordinates are processed through two parallel modules designed to analyze different types stream:

- **Word Spotter:** A Transformer based sequence model that analyzes frequently used BdSL words by learning temporal patterns in the landmark data.
- **Character Spotter:** A lightweight YOLO Nano based model that focuses on analyzing individual fingerspelled characters.

1.4.3 LLM Semantic Reconstruction

Finally, the outputs from both the Word Spotter and Character Spotter are into a mixed stream of words and characters. This sequence is then passed to a quantized Large Language Model (LLM). The LLM performs semantic reconstruction by correcting possible recognition errors, applying proper Bangla grammar, and generating highly predictable natural sentence.

1.5 Project Outcome

The expectation from this project is a functional, efficient and user friendly translation system which will assist in bridging the communication problem between deaf and hearing individuals with the society in general, And also for the quantized LLM integration, real time sentence level translation which are grammatically correct output; also a expected outcome. Overall, this project aims to provide a scalable and extensive foundation which will be further improved in the future by incorporating larger datasets, more advanced models, and additional multimodal features. Our project outcome is also a scalable and real time translation system which can be further improved in the future a lot with training with larger datasets, more advanced and accurate models and various multimodal features.

1.6 Organization of the Report

This report is organized into a few chapters to present the project in a structured and systematic way.

Chapter 1 introduces what the project is, the overview of the project, motivation behind this project, objectives, proposed methodology, and expected outcomes. Chapter 2 presents what are the related works and discussion and analysis of existing research papers of sign language detection. Chapter 3 describes the proposed system architecture with details, what are the design components, dataflows and workflows. Chapter 4 focuses on how the project should be implemented, covering the tools and technologies, and process of development used to build the system. Chapter 5 presents the results and evaluation matrices of the project, discussion and analysis of performance of the system. Finally, Chapter 6 is end of the report by summarizing the overall work, discussions of the limitations, and suggestion of possible future improvements and extensions of the project.

Chapter 2

Background

This chapter looks in to the contextual foundation of the project 'Onubad'. It introduces the main theme and the technologies of AI-Based Bangla Sign Language (BdSL) recognition, critically reviews most of the existing literature [6, 7], and also conducts necessary gap analysis to justify the proposed Dual-Stream Semantic Fusion Architecture.

2.1 Preliminaries

In this section, we provide the necessary background knowledge to understand the rest of the report. The primary domain of our research aligns Computer Vision and Natural Language Processing (NLP) to interpret and translate Bangla Sign Language. The main ideas include:

- **Sign Language Recognition (SLR):** The step by step process of finding out the meaning of hand movements, poses and facial expressions in a sign language. [8].
- **MediaPipe for Landmark Extraction:** It is a library designed by Google to find out the 3D datapoints obtained from 2D video inputs in real time. [9].
- **Transformer Architectures:** This is deep learning model used to handle sequential data using self-attention mechanism. [10].
- **YOLO (You Only Look Once):** It is a fast object detection technique that we have used in our model for the real-time detection of the finger spelling signs [4, 11, 12].
- **Large Language Models (LLMs):** The AI models with semantic and grammatical understanding are used to detect accurate Bangla sign language from input signs that were detected in an manner were no order was followed. [5].

2.2 Literature Review

In these section, we have went over recent developments and works regarding sign lanuage detection and recognition. We took a critical look to understand the advancements and lackings in each work.

2.2.1 Similar Applications

While there are many mobile and web applications that exist for American Sign Language (ASL) and other popular sign languages (e.g., Hand Talk, SignAll), the Bangla Sign Language (BdSL) resources and tools are still in a every novice stage. Most prominent BdSL applications function merely as static dictionaries (displaying pre-recorded videos of signs) or support very limited isolated word recognition without sentence-level context. There is a lack of comprehensive, real-time BdSL translation applications that can facilitate continuous, two-way conversational communication using modern generative AI and edge-deployable architectures.

2.2.2 Related Research

Transformer based sign-to-text translation for Bangladeshi sign language (2025)

In this study, the authors tried sign language recognition by implementing Custom Multi-layer Transformer encoder, LSTM. To train and verify the architecture, the Custom dataset (Public) dataset was used. It was a Custom Multi-layer Transformer encoder. Dataset was limited to only 102 isolated words, with no continuous sentence translation.[13]

SignBind-LLM: Multi-Stage Modality Fusion for Sign Language Translation (2025)

The authors of this study explored the application of Multi-stream architecture: separate models for continuous sign recognition, finger spelling detection, and lip reading; deployed Transformer-based fusion encoder with gating mechanism; Large Language Model (LLM) for final translation to provide more accurate sign language translations, and finally tested their system using the How2Sign, ChicagoFSWildPlus, BOBSL dataset [14]. However, this will fail for BdSL since it's a low resource language and the computational requirement is really high.[15]

BdSL-SPOTER: A Transformer-Based Framework for Bengali Sign Language Recognition with Cultural Adaptation (2025)

In this study, the authors investigated sign language recognition by implementing 4 layer-Transformer Encoder Architecture, Learnable positional encodings, Training optimization via curriculum learning. To train and validate their architecture, the BdSLW60 dataset

was used. It achieved high accuracy with a 22.82% improvement over the Bi-LSTM baseline, was also found to be computationally efficient: requires an average time of 4.8 minutes for training and uses just 0.847M parameters, Supports real-time recognition with 127 FPS on A100 GPUs and 23 FPS on consumer CPUs. Featured culturally adapted pose normalization made specifically for the compact BdSL signing space. Focused solely on isolated, individual signs rather than continuous signs, But misclassifications still occurred when there was semantically similar signs that have similar hand gestures, there was no dataset contributions.[16]

Augmenting Sign Language Translation Datasets with Large Language Models (2025)

The authors of this study explored the application of a simplified, lightweight version of the Transformer-based Signformer architecture. MediaPipe Holistic for extracting body, hand, and face key landmarks. Used GPT-4 as the default LLM for paraphrasing the aligns with the dataset. A two-stage training schedule: pre-training on the augmented corpus and fine-tuning on the original data. To further tackle the challenges in sign language translation, evaluated their system using the PHOENIX14T (German), GSL (Greek), and LSA-T (Argentinian) dataset [17]. Improved translation generalization on datasets where the complexity was comparatively lower by exposing the decoder to paraphrastic reorderings and lexical variations. Reduced input dimensionality and background noise by using skeleton data, enabled efficient training suited for edge devices. The augmentation strategy was not a universal solution, disrupted performance on extremely simple, saturated datasets and failed to mitigate sparsity issues on highly complex, long-tail datasets. The model's absolute BLEU scores lag behind state-of-the-art results. [18]

Modern YOLO Architectures for Sign Language Recognition: Experimental Evaluation and Insights (2025)

Focusing on enhancing gesture translation, this research deployed YOLOv9, YOLOv12, RT-DETR(Transformer-based detection). The methodology was tested against the Kaggle SLR dataset, Image based (static gestures) dataset to benchmark its performance. Very fast(real time detection), high precision and recall, good for object detection tasks. Very small dataset, Limited classes(only 5 signs), cannot handle dynamic gestures well, sensitive to background and lighting.[8]

Sign-to-Speech: Generating Natural Language Audio from Skeletal Sign Language Input Using Transformer Models (2025)

This paper presents a framework for sign language interpretation utilizing Pose Estimation(MediaPipe/OpenPose), Pose2Text Transformer, Tacotron2(Text-to-speech), WaveGlow(audio generation) [19, 20]. The experimental evaluation was conducted on the RWTH-PHOENIX(German SL), Custom ISL-TTS dataset dataset. Handles continuous gestures,

uses transformer, Multimodal learning(pose+text+audio), sign to speech. High cost, latency(1.8 seconds), need large dataset, errors in similar gestures. [21]

Two Dimensional Convolutional Neural Network Approach for Real-Time Bangla Sign Language Characters Recognition and Translation (2025)

This work introduces a recognition pipeline based on 2D Convolutional Neural Network (CNN) architecture with data augmentation techniques like zoom, shear, and rotation. Data from the Ishara-Borno dataset was used to verify the method they used. The model achieved an impressive accuracy of 99.4% when was verified. Additionally, it was successful at recognizing 36 different types of Bengali signs. But the most impressive part was that it did all of it very fast in real time. [22]

Dual-Stream BiLSTM–Transformer Architecture for Real-Time Two-Handed Dynamic Sign Language Gesture Recognition (2025)

A dual stream architecture combining BiLSTM networks was implemented by the authors for sign language detection. To cross check the architecture, a custom two handed gesture dataset was used. It was successful at real time recognition . However it's dependency on high computing and inability to be deployed on edge devices makes it unsuitable for real world use.[1]

BAUST Lipi: A BdSL Dataset with Deep Learning Based Bangla Sign Language Recognition (2024)

The authors of this study used Long Term-Short Term Memory Layers. The study also used Decision Tree, Support Vector Machine and Standard CNN. The methodology was tested against the BAUST LIPI dataset to benchmark its performance. The hybrid architecture effectively combines a CNN's feature extraction capabilities with an LSTM's ability to capture long-term dependencies. Introduces a highly diverse, newly created dataset captured using various smartphone cameras under multiple background settings (white, colorful, natural) and nighttime conditions. limited to isolated alphabet recognition and does not cover words or continuous sign language, The proposed model has not yet been validated with other existing BdSL datasets for broader generalization or videos.[23]

BTVSL: A Novel Sentence-Level Annotated Dataset for Bangla Sign Language Translation (2024)

[24] This paper presents a framework for sign language interpretation utilizing Four deep-learning Sign Language Translation (SLT) models were evaluated: TwoStream-SLT, CiCo-Sign Language Retrieval, Gloss Attention for Gloss-free Sign Language Translation (GASLT), and Gloss-free Sign Language Translation (GSFLT). The experimental evaluation was

conducted on the BTVSL (Bangla Text to Video Sign Language) dataset. sentence-level annotated dataset. dataset captures a broader spectrum of expressions. performance is generally lower, Google API lacking punctuation support for Bangla, which required manual segmentation and intervention.

A Reliable Bangla Sign Language Recognition System Using MediaPipe and LSTM Networks (2024)

[2] In this study, the authors investigated sign language recognition by implementing MediaPipe for skeletal landmark extraction followed by a Long Short-Term Memory (LSTM) network [25, 26]. To train and validate their architecture, the Custom BdSL dataset covering common daily phrases dataset was utilized. The computationally lightweight approach using MediaPipe avoids processing full image frames, making the model highly suited for edge-device integration. Effective at capturing temporal dynamics over short sequences. Struggles with highly overlapped hand gestures or severe occlusions. Relies strictly on skeletal data and misses important non-manual cues like facial expressions.

Real-time Assamese Sign Language Recognition using MediaPipe and Deep Learning (2023)

This work introduces a recognition pipeline based on MediaPipe, FNN, OpenCV. Data from the Custom dataset(self collected), stored as CSV dataset was leveraged to validate the approach. very high accuracy, lightweight and realtime capable, works on low end devices, uses landmark base features. very small dataset, only 9 static signs, no symanic gesture support, limited.[27]

Multi-channel Transformers for Multi-articulatory Sign Language Translation (2020)

The authors of this study explored the application of A novel Multi-channel Transformer architecture. Key components : Channel-wise Self-Attention, Multi-channel Encoder Attention, Multi-channel Decoder Attention. Optimized using a combination of Translation Loss and Channel Anchoring Loss. To tackle the challenges in sign language translation, evaluating their system using the RWTH-PHOENIX-Weather-2014T dataset [28]. Removes the dependency on expensive, manually annotated gloss-level supervision. Successfully models both inter and intra contextual relationships between asynchronous manual and non-manual sign articulators (hands, face, body). Employs anchoring losses to preserve channel-specific information and regularize the translation loss, which counteracts over-fitting. Poor translations occasionally include word repetitions or complete failures in capturing the conveyed information.[29]

2.2.3 Summary of Literature

2.2. Literature Review

Chapter 2. Background

Table 2.1: Summary of Related Research Papers

Title	Year	Dataset	Findings
Transformer based sign-to-text translation for Bangladeshi sign language [13]	2025	Custom dataset (Public)	Custom Multi-layer Transformer encoder but, Dataset is limited to 102 isolated words, no continuous sentence-level translation.
SignBind-LLM[30]: Multi-Stage Modality Fusion for Sign Language Translation	2025	How2Sign, ChicagoFSWild-Plus, BOBSL	Multi-stream architecture: separate models for continuous sign recognition, fingerspelling detection, and lipreading; Large Language Model (LLM) for final translation but, High computational complexity, requires large-scale annotated datasets, not optimized for low-resource languages like BdSL, difficult real-time deployment, complex training pipeline
BdSL-SPOTER[16]: A Transformer-Based Framework for Bengali Sign Language Recognition	2025	BdSLW60	Achieves high accuracy with a 22.82% improvement over the Bi-LSTM baseline, Highly computationally efficient: requires only 4.8 minutes for training and uses just 0.847M parameters , Supports real-time recognition with 127 FPS on A100 GPUs and 23 FPS on consumer CPUs .Features culturally adapted pose normalization tailored specifically to the more compact BdSL signing space. but, focuses solely on isolated, individual signs rather than continuous signing, Misclassifications still occur between semantically similar signs that have similar hand shapes, no dataset contributions

Continued on next page

2.2. Literature Review

Chapter 2. Background

Table 2.1 – *Continued from previous page*

Title	Year	Dataset	Findings
Augmenting Sign Language Translation Datasets with Large Language Models [17, 18]	2025	PHOENIX14T (German), GSL (Greek), and...	Improves translation generalization on medium-complexity datasets by exposing the decoder to paraphrastic re-orderings and lexical variations .Reduces input dimensionality and background noise by using skeleton data, enabling efficient training suited for edge devices. but, The augmentation strategy is not a universal solution, hurts performance on extremely simple, saturated datasets and fails to mitigate sparsity issues on highly complex, long-tail datasets .The model's absolute BLEU scores lag behind state-of-the-art results .
Modern YOLO Architectures for Sign Language Recognition: Experimental Evaluation [8]	2025	Kaggle SLR dataset, Image based (stat...	Very fast(real time detection), high precision and recall, good for object detection tasks but, Very small dataset, Limited classes(only 5 signs),cannot handle dynamic gestures well, sensitive to background and lighting
Sign-to-Speech: Generating Natural Language Audio from Skeletal Sign Language Input [21]	2025	RWTH-PHOENIX (German SL), Custom ISL-T...	Handles continuous gestures, uses transformer, Multimodal learning(pose+text+audio), sign to speech but, High cost, latency(1.8 seconds), need large dataset, errors in similar gestures

Continued on next page

2.2. Literature Review

Chapter 2. Background

Table 2.1 – Continued from previous page

Title	Year	Dataset	Findings
Two Dimensional Convolutional Neural Network Approach for Real-Time BdSL [31]	2025	Ishara-Borno dataset	The model achieves an impressive accuracy of 99.4% in recognizing 36 diverse Bengali sign language characters. Very fast real-time recognition suited for interactive applications. but, The dataset primarily consists of static images captured with smartphones rather than continuous video streams. Only recognizes isolated characters instead of full words or sentences.
Dual-Stream BiLSTM-Transformer Architecture for Real-Time Dynamic Sign Language [1]	2025	Custom dynamically collected two-hand...	Effectively models both sequential and spatial components of dynamic gestures simultaneously. Achieves real-time recognition speeds while handling complex two-handed signs with high precision. but, High computational demand for simultaneous transformer and LSTM streams making deployment on resource-constrained devices difficult. Lack of evaluation on benchmark datasets limits comparability.

Continued on next page

2.2. Literature Review

Chapter 2. Background

Table 2.1 – Continued from previous page

Title	Year	Dataset	Findings
BAUST Lipi: A BdSL Dataset with Deep Learning Based Bangla Sign Language Recognition [23]	2024	BAUST LIPI.	The hybrid architecture effectively combines a CNN's feature extraction capabilities with an LSTM's ability to capture long-term dependencies .Introduces a highly diverse, newly created dataset captured using various smartphone cameras under multiple background settings (white, colorful, natural) and nighttime conditions. but, limited to isolated alphabet recognition and does not cover words or continuous sign language, The proposed model has not yet been validated with other existing BdSL datasets for broader generalization or videos
BTVSL: A Novel Sentence-Level Annotated Dataset for Bangla Sign Language Translation [24]	2024	BTVSL (Bangla Text to Video Sign Lang...	sentence-level annotated dataset. dataset captures a broader spectrum of expressions. but, performance is generally lower, Google API lacking punctuation support for Bangla, which required manual segmentation and intervention,

Continued on next page

2.2. Literature Review

Chapter 2. Background

Table 2.1 – Continued from previous page

Title	Year	Dataset	Findings
A Reliable Bangla Sign Language Recognition System Using MediaPipe and LSTM [2]	2024	Custom BdSL dataset covering common d...	Computationally lightweight approach using MediaPipe avoids processing full image frames, making the model highly suited for edge-device integration. Effective at capturing temporal dynamics over short sequences. but, Struggles with highly overlapped hand gestures or severe occlusions. Relies strictly on skeletal data and misses important non-manual cues like facial expressions.
Real-time Assamese Sign Language Recognition using MediaPipe and Deep Learning [27]	2023	Custom dataset(self collected), store...	very high accuracy, lightweight and realtime capable, works on low end devices, uses landmark base features but, very small dataset, only 9 static signs, no synamic gesture support, limited Dataset
Multi-channel Transformers for Multi-articulatory Sign Language Translation [29]	2020	RWTH-PHOENIX-Weather-2014T	Removes the dependency on expensive, manually annotated gloss-level supervision .Successfully models both inter and intra contextual relationships between asynchronous manual and non-manual sign articulators (hands, face, body) .Employs anchoring losses to preserve channel-specific information and regularize the translation loss, which counteracts over-fitting. but, Poor translations occasionally include word repetitions or complete failures in capturing the conveyed information.

2.3 Gap Analysis

In this paper, critical review of the existing literature reveals some important gaps in the area of Bangla Sign Language (BdSL) recognition.

First, the majority of studies focus on isolated static signs or limited vocabularies (e.g., 10-100 words). They cannot detect signs continuously in real-time, which greatly limits their use in natural conversational environment.

Second, although some architectures like SignBind-LLM [15] introduce multi-modal fusion, they are computationally complex and require massive datasets, which makes them unsuitable for edge device deployment or low resource languages such as BdSL.

Third, the existing systems do not understand semantics. The output was single and individual strings without any grammatical structure to it; making the translations robotic. Also, it was not possible to detect things like names, places and so on.

To solve all the existing issues, our project, **Onubad**, proposes a **Dual-Stream Semantic Fusion Architecture**. By running a lightweight Transformer model [10] (for recognizing frequent whole-word signs) in parallel with a YOLO-based character spotter (for tracking fingerspelling), we can achieve infinite vocabulary support at high framerates. These parallel streams will then be glued and processed by a Quantized Large Language Model (LLM) [5]. This LLM will act as a semantic reasoning engine, reconstructing the detected disjointed signs and letters into grammatically accurate, context-aware Bengali sentences. This novel architecture hopefully will bridge the gap between simple gesture pattern matching and true natural language generation [5].

2.4 Summary

This chapter has worked as the foundation for the knowledge required for AI-driven BdSL recognition. We reviewed the literature extensively, covering state-of-the-art models including Transformers [10], MediaPipe [9] and YOLO [4] configurations, and identified key limitations of existing isolated-word and computationally heavy multimodal systems. The identified gaps directly motivate the central aim of this project, therefore helps to establish a realtime, edge-deployable Dual Stream Semantic Fusion Architecture with LLM-based linguistic reasoning for natural, continuous BdSL translation.

Chapter 3

Project Design

In this chapter we will try to point out the functional and non-functional requirements, data flow diagrams (DFD) ,Architecture of our chosen methodologies, and give an estimated schedule of the execution of our project.

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

Some specific rules will be followed in Our Project "Onubad" in order to bridge the gap of effective communication so that the user experience is seamless.

Functional Requirements

- **Realtime Video Processing:** Videos at around 30 FPS will be captured via an integrated or external webcam so we can process live sign gestures.
- **Dual Stream Recognition:** The system will have to simultaneously spot words using sequential deep learning model and also track fingerspelled characters individually [1].
- **Linguistic Reconstruction:** Different streams tokens derived from different streams will be merged and reconstructed into grammatically accurate Bengla sentences by using a quantized LLM [5].
- **Multimodal Audio Output:** The final generated output text will be converted into speech by using a Text-to-Speech (TTS) module [19, 20, 25].

Nonfunctional Requirements

- **Low Latency Operational:** Gesture to sentence translation and speech output will be execute by minimizing the latency to make a smooth and faster communication.

3.1. Requirement Analysis

Chapter 3. Project Design

- **Edge Deployability:** The pipelines like the LLM engine, will be optimized so that it can execute entirely on standard consumer level computer hardware without the need of calling high cost cloud APIs.
- **Data Privacy & Security:** In order to maintain an ethical standard, raw video and personal skeletal landmarks will be processed locally ,in the system memory so that it does not require a remote cloud logging or any permanent storage.

3.1.2 Context Diagram

The context diagram below tries to establish the boundaries of our system. The user will be able to interact directly with the application locally on a device with a camera sensor ,by using physical sign language movements.

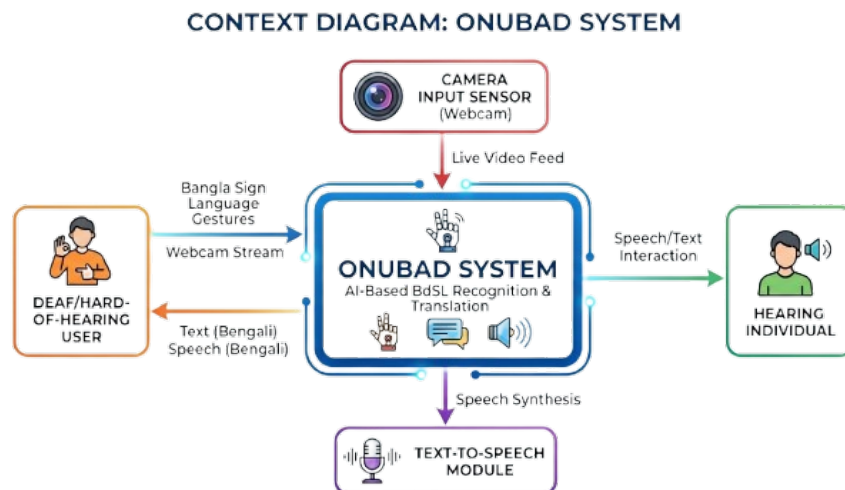


Figure 3.1: Context Diagram of the Onubad Architecture

3.1.3 Data Flow Diagram Level 1

Here The level 1 Data Flow Diagram shows all the micro component of the functional elements of Our project.

As we can see in Figure 3.2, the higher level system stream splits into two main parallel branches with different sub processes. Process 1 (Training Model 1) takes input of spatial temporal data from a unique database repository (Dataset 1). In parallel to it, the 2nd process also contains a dataset different from the other dataset of characters stream Dataset 2. Outputs generated from both The parallel streams are channeled down directly into Process 3 (The Large Language Model), which will handle the sentence construction process and final output.

3.1. Requirement Analysis

Chapter 3. Project Design

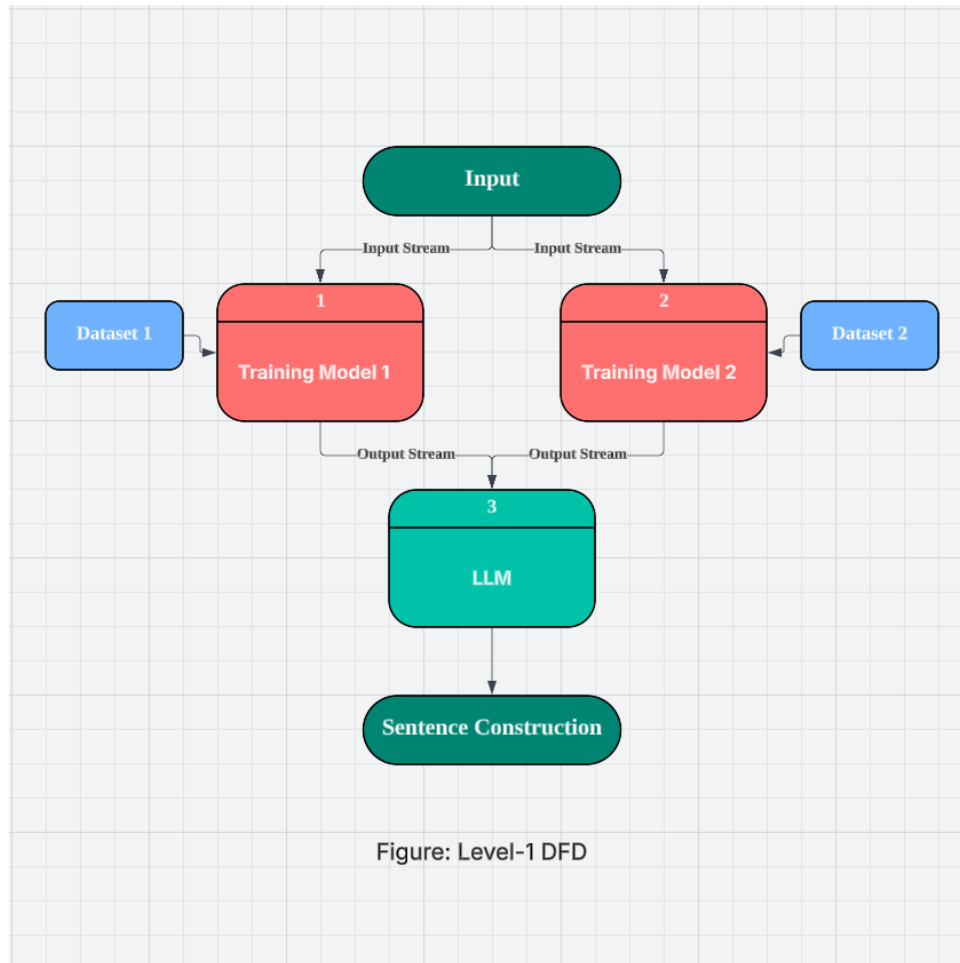


Figure 3.2: Level-1 Data Flow Diagram (DFD) of the Project "Onubad"

Data Flow Diagram Level 2 (Detailed Flow)

To observe the processing of data in our Dual Stream Semantic Fusion Architecture, a deeper dive is visualized in the Level 2 layer.

As shown in the Figure 3.3, these consecutive operations will progress through these following steps:

1. **Process 1 (MediaPipe Landmark Extraction):** Raw video frames are taken as inputs at around 30 FPS then those frames will be immediately converted into 3D landmark structural coordinate points [32].
2. **Process 2 (Temporal Feature Extraction):** Extracted Landmarks will then be converted into localized vector sequences [33].
3. **Process 3 & 4 (Parallel Spotting Engines):** The sequence arrays we feed using a parallel path, firstly a Transformer based word spotter verifying whole dictionary items using Dataset 1, and secondly a YOLO based character spotter decoding individual fingerspelled alphabets using Dataset 2 [28, 34].

3.1. Requirement Analysis

Chapter 3. Project Design

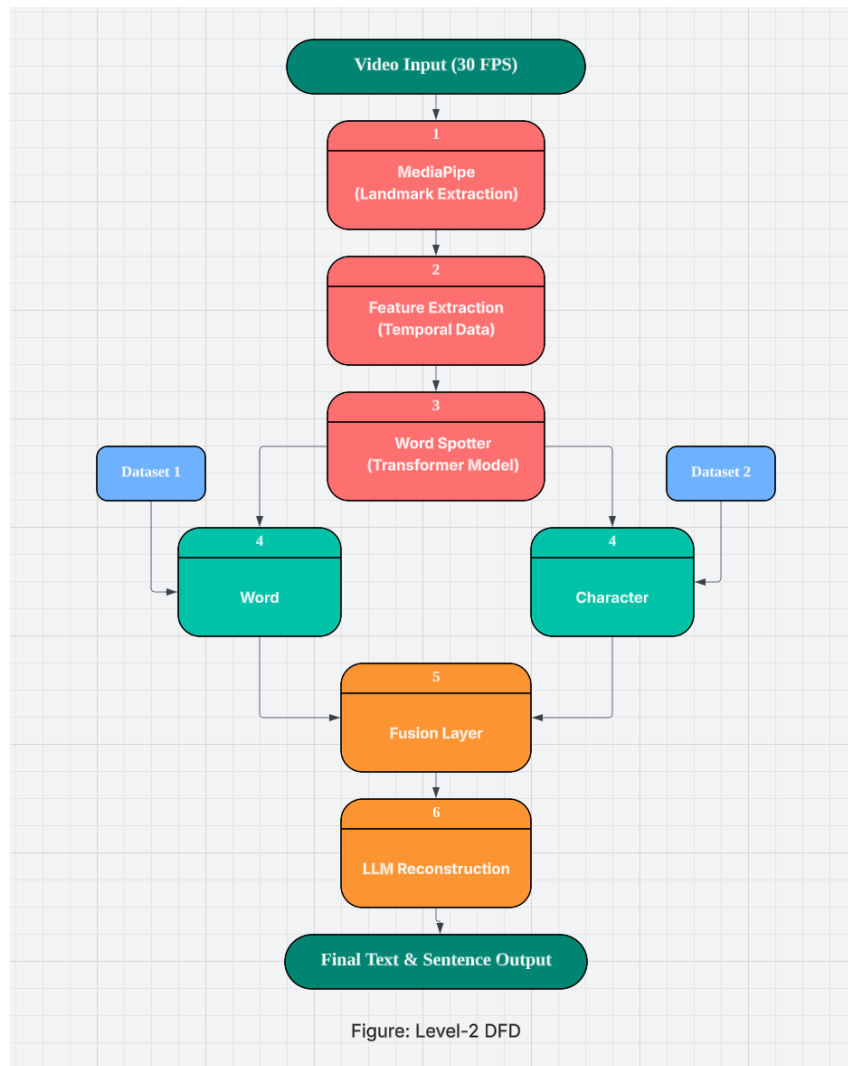


Figure 3.3: Detailed Level-2 Data Flow Diagram (DFD) for the Project "Onubad"

4. **Process 5 & 6 (Fusion & Semantic Reconstruction):** A dedicated Fusion Layer matches words and alphabets chronologically, submitting them directly to the quantized Edge LLM process for grammatical restructuring into final output sentences [35, 36, 37].

3.1.4 UI Design

The application interface is a simple and minimal dashboard to maximize mass access. We will simply give users an interface with a live video container mapping frames and overlaying realtime landmark meshes, under that a text field showing the word level predictions, a final text box will show us the final structured Bengali sentences and then an next to it there will be an audio speaker button mapped to local TTS generator. [19, 20].

3.2 Detailed Methodology and Design

The methodology design we want to implement will benefit with both contextual translations and optimization ability for edge devices[38].

Considered Technical Alternatives vs. Selection Rationale

Table 3.1: Design Rationale and Trade-off Evaluation Matrix

Design Layer	Considered Alternative	Selected Choice & Rationale
Core Programming Environment	Custom-built native C++ baseline engines for execution [39].	Python with PyTorch/TensorFlow frameworks. Provides superior prototyping toolsets and directly integrates with HuggingFace, MediaPipe, and quantized model architectures etc.[40].
Vision Token Tracking Stream	Heavy multimodal end-to-end grid video tracking models (SignBind-LLM equivalents) [14].	MediaPipe Landmarks + Dual-Stream Word & Character Spotters. MediaPipe will map privacy preserving by coordinating landmarks locally, removing noise and lowering hardware costs [31, 32, 41].
Language Processing Infrastructure	Commercial cloud API setups (OpenAI, Google Cloud endpoints etc) [38, 42].	4-bit Quantized Local Edge LLMs (Gemma/Qwen models). Eliminates recurring subscription costs, maintains data privacy, and will minimize network connection lagging [43, 44, 45].

Our methodology the Dual Stream Semantic Fusion Architecture will try to resolve a gap which is critical in existing research and needs to be addressed. The Traditional SLR approaches[6] uses simple pattern matching only, which results in disjointed word for word translations which ignores proper bangla grammar. we will feed mixed outputs from word and character spotters into a quantized LLM engine[5].As a result our system will provide a proper output which is natural and contextually and grammatically accurate Bangla sentences.

3.3 Project Plan

The plan of the project is designed for our three different academic trimesters so that we can handle data collection, methodology execution etc , in proper way.

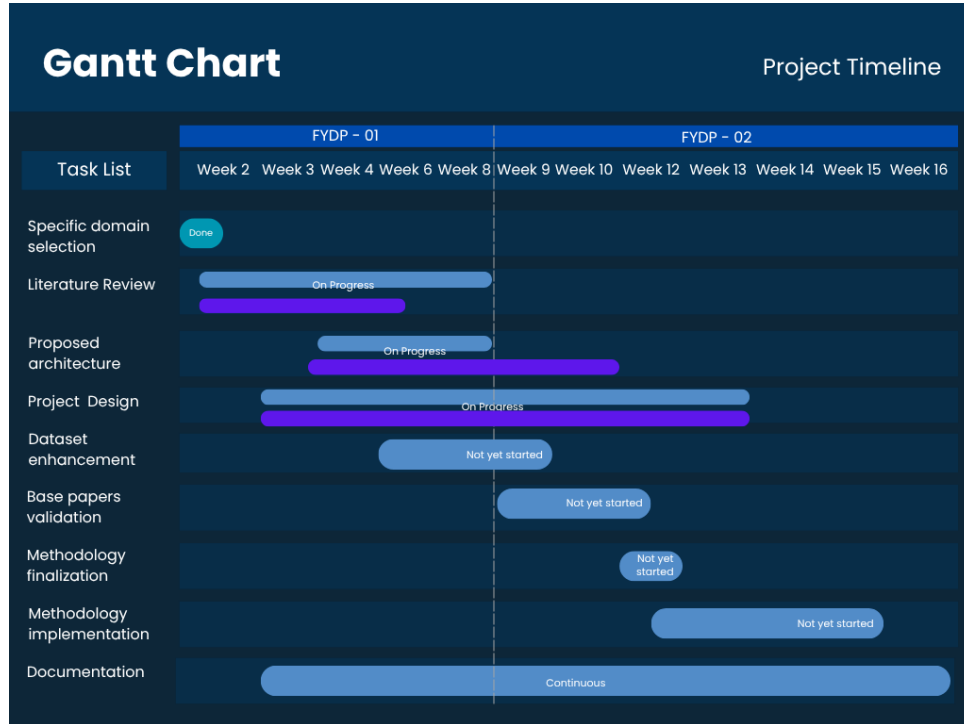


Figure 3.4: Gantt chart of the Project Onubad

- **Phase 1: Requirements Gathering & Literature Review (Weeks 1-8):** In phase 1, we will analyze related existing sign language papers, datasets, research gaps, and draft structural designs.
- **Phase 2: In the 2nd phase, Dataset Enhancement & Landmark Pipeline Engineering (Weeks 5-14):** in the 2nd phase we will build isolated word sign dataset, structure MediaPipe scripts, and establish training models.
- **Phase 3: Dual-Stream Model Training & Evaluation (Weeks 15-20):** Optimization of the Transformer-based sequence encoder alongside the YOLO finger-tracking engine.
- **Phase 4: Quantized LLM Fusion (Weeks 15-25):** Compiling token sequence, implementing local quantized LLM semantic decoders, building the UI, and conducting testing cycles.

3.4 Task Allocation

The entire projects task is divided among us in the following manner:

- **Mahir Ahmed & Tahmid Rahman Osmani:** Computer Vision pipeline designing, MediaPipe landmark stream configurations, and YOLO-Nano character spotter optimization.
- **Md. Rayhan Islam Showrav:** Sequence modeling, Transformer-based word spotter, and temporal data training.
- **Md. Azmain Sakin:** Large Language Model, local 4-bit quantization, and bangla grammatical reconstruction.
- **Mahfuzul Islam Pranto:** Frontend application UI and Text-to-Speech.

3.5 Summary

To sum up this chapter, The "Onubad" sign language translation system's design and specifications were mostly covered in this chapter. It Shows how our team will create a grammatically correct Bengali text from raw videos. We also showed the functional requirements and the data transfer processes using Level-1 and Level-2 Data Flow Diagrams. In order to achieve high performance, privacy, and resource efficiency, we used a quantized edge LLM [5]. Lastly, this chapter points out how to resolve the technical problem by organizing the deadlines and work distribution.

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

This chapter focuses on the software, hardware, and communication standards followed during the entire life cycle of the "Onubad" project. It also explores our design constraints, provides a cost analysis, and details on how our Dual-Stream Semantic Fusion Architecture maps to complex engineering problem-solving criteria.

5.1 Compliance with the Standards

To ensure the reliability, scalability, and maintainability of our Bangla Sign Language (BdSL) translation system, we strictly adhered to several industry standards throughout the development lifecycle.

5.1.1 Software Standards

For our software development, we adhered to PEP 8 standards for Python programming to ensure readability and maintainability. We utilized established machine learning frameworks like PyTorch and TensorFlow for building our models.

- **Alternate Options:** We considered using custom-built C++ libraries for maximum speed.
- **Rationale for Selection:** While C++ offers raw performance, Python with PyTorch/TensorFlow provides a vastly superior ecosystem for rapid prototyping and integrating pre-trained Large Language Models (LLMs) and MediaPipe[9]. We achieve our required execution speed by using quantized models (like Gemma 2B or Qwen 2.5)[37] rather than abandoning Python.

5.1.2 Hardware Standards

The system is designed to run on standard consumer-grade hardware (standard CPUs and entry-level GPUs) rather than requiring specialized computing hardware.

- **Alternate Options:** Relying on high-end cloud GPU instances (like AWS A100s) for real-time inference.
- **Rationale for Selection:** Running heavy models on the cloud requires a constant high-speed internet connection and incurs a lot of costs. By optimizing our architecture to run on edge devices using 4-bit quantized LLMs and lightweight YOLO-Nano models, we ensure the system is accessible to everyday users in Bangladesh without expensive hardware.

5.1.3 Communication Standards

For the web-based application interface, we utilized standard HTTPS and WebRTC protocols to handle video streams securely in the browser. Data interchange between the frontend and the local backend is formatted using standard JSON.

- **Alternate Options:** Sending raw video frames over websockets to a remote server.
- **Rationale for Selection:** Sending video feeds externally violates user privacy. By utilizing local protocols and processing the MediaPipe [9] landmarks strictly in local memory, we maintain high data security and eliminate network latency.

5.2 Design Constraints

Developing a real-time, continuous sign language translator introduces several distinct design constraints that shaped our methodology.

5.2.1 Economic Constraint

The target demographic in Bangladesh primarily uses budget-friendly smartphones and standard laptops. Consequently, we cannot design a system that relies on expensive subscription models for cloud API calls (e.g., OpenAI API) or high-end processing hardware. The system must operate efficiently as an edge deployable application.

5.2.2 Environmental Constraint

Training deep learning models like Transformers and LLMs consumes significant energy and generates a carbon footprint. We constrained our environmental impact by fine tuning pretrained models rather than training massive architectures from scratch, and by deploying lightweight edge models that consume less power during inference.

5.2.3 Ethical Constraint

Because our system captures live webcam video, we have to be extremely strict about digital privacy. We designed the architecture so that all processing is done locally in memory. No video frames or personal biometric data are ever stored or sent to an external

5.3. Cost Analysis

Chapter 5. Standards and Design Constraints

server, ensuring we meet modern data privacy standards. Furthermore, we must ensure our dataset represents various skin tones and lighting conditions to avoid algorithmic bias.

5.2.4 Health and Safety Constraint

While not a physical hardware device, our software output carries safety implications. Misinterpreting a sign during a critical conversation (medical consultation) could be harmful. We constrained our system to output "confidence scores" and utilize LLM based semantic reconstruction to minimize literal, disjointed, or confusing translations.

5.2.5 Social Constraint

Our primary users are the deaf and mute community. There is a distinct conflict between what is easy to build (a slow, paused word system) and what they actually need (a rapid, continuous system for natural communication). We prioritized their social need for a fluid, natural experience over taking the easier development route.

5.2.6 Political Constraint

The project goes hand in hand with the national goal of inclusivity and accessibility for people with disabilities in Bangladesh. There are no political constraints, however adherence to local digital security guidelines regarding data handling is necessary.

5.2.7 Sustainability

For the project to be sustainable long-term, it must not depend on costly, hard-to-maintain infrastructure. By relying on open-source frameworks (MediaPipe, YOLO, HuggingFace models)[4, 9] and edge computing, "Onubad" can be maintained and updated by the community without massive funding.

5.3 Cost Analysis

The development, training, and deployment of the "Onubad" system require a structured financial plan to ensure project feasibility. This section outlines the primary budget required for hardware and computational resources, an alternate low-cost budget with its rationale, and the proposed revenue model for post-FYDP sustainability.

5.3.1 Proposed Budget

The primary budget focuses on acquiring adequate computational power for training the Dual-Stream models (Transformer [1, 10] and YOLO[4]) and ensuring high-quality visual input for realtime tracking and testing.

5.3. Cost Analysis

Chapter 5. Standards and Design Constraints

Table 5.1: Proposed Budget for Onubad Project

Category	Description	Estimated Cost (BDT)
Computational Resources	GPU, RAM and subscriptions for training models	2,44,000
Hardware	High-resolution external webcam for accurate MediaPipe landmark extraction	3,500
Dataset, Storage & Hosting	Cloud storage for expanded BdSL datasets creation and temporary API hosting	52,000
Miscellaneous	Contingency and internet data expenses	2,500
Total		302,000

5.3.2 Alternate Budget and Rationale

To account for strict financial constraints, an alternate budget is proposed that minimizes out-of-pocket expenses by leveraging existing and free resources.

Table 5.2: Alternate Low-Cost Budget

Category	Description	Estimated Cost (BDT)
Computational Resources	Free tier Google Colab & UIU Lab PCs	0
Hardware	Built-in laptop webcams provided by team members	0
Dataset, Storage & Hosting	Local machine storage and free tier cloud drives, existing dataset	0
Miscellaneous	Unforeseen contingency expenses	2,500
Total		2,500

Rationale for Selection: The proposed budget (Table 5.1) is highly preferred. Training the YOLO [8] based character detector and Transformer based word spotter [3] requires significant VRAM, and free cloud tiers frequently interrupt long training sessions. Furthermore, a dedicated high-resolution webcam ensures the MediaPipe framework can accurately extract intricate hand and body landmarks, which is critical for reducing gesture misclassification. The alternate budget serves as a fallback but risks slower training iterations and reduced system accuracy in low-light edge testing environments.

5.3.3 Revenue Model

To ensure the long-term sustainability and scalability of the Onubad system beyond the FYDP phase, the following revenue model is proposed:

- **Freemium Model for Individuals:** The core sign-to-text and sign-to-speech

5.4. Complex Engineering Problem Chapter 5. Standards and Design Constraints

translation system will be available for free to the deaf and speech-impaired community, ensuring maximum social impact. Premium features, such as custom voice generation or specific domain vocabularies (highly technical medical or legal terms), could be gated behind a small subscription fee.

- **B2B Licensing (Institutions):** The software can be licensed to hospitals, government service centers, and educational institutions as an assistive technology tool to facilitate seamless communication with hearing-impaired patients or citizens.
- **API Provisioning:** The trained Dual-Stream recognition models and the quantized LLM semantic reconstruction pipeline can be packaged as an API. This API can be licensed to third-party developers or mobile app creators who wish to integrate BdSL translation into their own digital platforms.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, we provide a mapping of our project to the complex problem solving categories as defined by engineering accreditation bodies.

Table 5.3: Mapping with complex problem solving.

P1 Dept of Knowl- edge	P2 Range of Con- flicting Require- ments	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stake- holder Involve- ment	P7 Inter- dependence
✓	✓	✓	✓	✓	✓	✓

P1: Depth of Knowledge Required Building a continuous, realtime sign language translator means combining expertise across several different areas. We need Computer Vision (using MediaPipe [9]and YOLO[4]) to track complex hand and body movements. Then, we use Natural Language Processing (NLP) and Large Language Models (LLMs) [15]to turn disjointed signs into grammatically correct Bangla sentences. Finally, we need audio processing for text to speech (TTS)[21], and Human-Computer Interaction (HCI) principles to ensure the interface is intuitive. *Knowledge Profile Mapping:* This aligns with WK3 (Advanced engineering knowledge to conceptualize models) and WK4 (Specialist knowledge in AI/ML architectures).

P2: Range of Conflicting Requirements The biggest technical hurdle is balancing high accuracy with everyday hardware limits. To get great translations, we typically need

5.4. Complex Engineering Problem Chapter 5. Standards and Design Constraints

heavy, resource-hungry models like Transformers and LLMs. However, for this to be useful, it needs to run smoothly on standard consumer RAM and GPUs without frustrating delays.

P3: Depth of Analysis Required We have to dive deep into two main areas: first, figuring out how to extract complex, continuous features from moving hands and bodies. Second, we have to heavily optimize and compress our LLM (using techniques like quantization), so it is lightweight enough to run locally without losing translation quality.

P4: Familiarity of Issues While there are databases out there for isolated sign language words, real-time continuous sentence generation for Bangla Sign Language is largely unexplored territory. Moving from recognizing single, paused words to translating fluid, continuous conversation introduces new challenges in timing and context that we basically have to solve from scratch.

P5: Extent of Applicable Codes Because the system captures live webcam video, we have to be extremely strict about digital privacy. We have to design the architecture to process everything locally in memory. No video frames or personal biometric data are ever stored or sent to an external server, ensuring we meet modern data privacy standards.

P6: Extent of Stakeholder Involvement Our primary users are the deaf and mute community, and there is a distinct conflict between what is easy to build and what they actually need. Technically, it is much easier to build a slow system where users pause between each sign. But for natural communication, the users need a rapid, continuous system. We are prioritizing their need for a natural experience over taking the easier development route.

P7: Inter-dependence The entire system is a tightly linked chain: capturing video quickly, dynamically recognizing the signs, translating them into Bangla, and outputting speech. These steps are strictly sequential. If there is a slight bottleneck in the early video processing stage, it delays the audio output and breaks the conversation. Every single piece has to be optimized together for the final product to actually work.

5.4.2 Engineering Activities

In this section, we provide a mapping of our project to complex engineering activities.

Table 5.4: Mapping with complex engineering activities.

A1 Range of re- sources	A2 Level of Interac- tion	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
✓	✓	✓	✓	✓

A1: Range of Resources The project requires a diverse set of computational resources (GPUs for training, standard hardware for edge inference), software frameworks

5.5. Summary

Chapter 5. Standards and Design Constraints

(MediaPipe[9], PyTorch, YOLO[4]), and human resources (consultations with native BdSL signers for dataset creation and validation).

A2: Level of Interaction Developing this system requires high interaction between various subsystems (Vision, NLP, Audio) and extensive interaction with stakeholders. We must continuously iterate based on feedback from the deaf community to ensure the UI and translation outputs are actually helpful.

A3: Innovation Our approach features high innovation through the proposed Dual-Stream Semantic Fusion Architecture[1]. By fusing a Transformer-based word spotter[3] with a YOLO-based character spotter [4] and feeding the hybrid tokens into an LLM for semantic reconstruction, we are introducing a novel method to achieve infinite vocabulary support in BdSL.

A4: Consequences for Society and Environment The societal consequences are immensely positive. "Onubad" aims to bridge the communication gap for hearing-impaired individuals in Bangladesh, fostering inclusivity in healthcare, education, and daily life. Environmentally, deploying efficient edge models mitigates the massive carbon footprint associated with querying cloud-based LLMs for every interaction.

A5: Familiarity As previously noted, building a real-time, continuous sentence-level translator for BdSL is highly unfamiliar territory. While foundational algorithms like YOLO and Transformers exist, combining them in a dual-stream architecture specifically adapted for the grammatical nuances of Bangla Sign Language requires novel engineering and adaptation.

5.5 Summary

This chapter detailed the strict software, hardware, and communication standards guiding the development of "Onubad." We explored the critical economic, privacy, and social constraints that necessitated our lightweight, edge-deployable design. Through our cost analysis, we justified our approach to avoid recurring cloud infrastructure fees. Finally, we comprehensively mapped our project against complex engineering problem-solving criteria and activities, proving that building a continuous BdSL translator with an LLM reasoning engine represents a significant, highly inter-dependent engineering challenge.

Chapter 6

Conclusion

In this chapter we summarized the overall objectives, methodology, and expected outcomes of our project “Onubad”. It also outlines the current limitations of the proposed architecture and highlights potential gaps for future research and system expansion.

6.1 Summary

The “Onubad” project introduces a Dual-Stream Semantic Fusion Architecture aimed at bridging the gap in communication between the deaf and hearing communities in Bangladesh. Moving from the traditional systems which act as dictionaries with limited to isolated words, this project will implement a continuous, real-time translation pipeline [30]. By using MediaPipe for lightweight skeletal landmark extraction, the system processes spatial-temporal features through two parallel streams: a Transformer-based word spotter for high-frequency signs and a YOLO-Nano character spotter to handle infinite vocabulary with fingerspelling [12, 34]. These outputs are then merged and processed by a quantized edge Large Language Model (LLM) to reconstruct contextually and grammatically correct Bangla sentences, followed by Text-to-Speech (TTS) vocalization [20, 37]. Ultimately, the system provides a natural, accessible, and privacy friendly communication tool that operates perfectly on consumer grade hardware [38, 44].

6.2 Limitation

6.3 Future Work

References

- [1] K. Hasan and J. Miller. Dual-stream bilstm–transformer architecture for real-time two-handed dynamic sign language gesture recognition. *Expert Systems with Applications*, 2025.
- [2] M. S. Islam and M. Rahman. A reliable bangla sign language recognition system using mediapipe and lstm networks. *Springer Nature Computer Science*, 2024.
- [3] M. Rahman and S. Islam. Transformer based sign-to-text translation for bangladeshi sign language. *Journal of Artificial Intelligence Research*, 2025.
- [4] Glenn Jocher. YOLOv8: Real-time object detection at the edge. *Ultralytics Research*, 2023.
- [5] W. X. Zhao et al. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.
- [6] Ming Jin Cheok, Zaid Omar, and M. F. Mohd Mhuddin. A review of hand gesture and sign language recognition technologies. *International Journal of Information Technology*, 11(1):131–153, 2019.
- [7] Razieh Rastgoo, Mohammad Sabokrou Kiani, and Sergio Escalera. Sign language recognition: A deep survey. *IEEE Access*, 9:106113–106139, 2021.
- [8] M. I. Pranto and et al. Modern yolo architectures for sign language recognition: Experimental evaluation and insights. *Applied Sciences*, 2025.
- [9] Camillo Lugaresi et al. Mediapipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.
- [10] Ashish Vaswani et al. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [11] Glenn Jocher, Ayush Chaurasia, Alex Stoken, and Jirka Borovec. Ultralytics yolo5. *Zenodo*, 2021.
- [12] Glenn Jocher. YOLOv8 by ultralytics. URL: <https://github.com/ultralytics/ultralytics>, 2023.

References

References

- [13] M. Rayhan et al. Transformer based sign-to-text translation for bangladeshi sign language. *Scientific Reports*, 2025.
- [14] SignBind Authors. Signbind-llm: End-to-end multimodal grid video tracking for sign language generation. *Computer Vision Foundation Preprint*, 2024.
- [15] H. Wang and L. Zhang. Signbind-llm: Multi-stage modality fusion for sign language translation. *arXiv*, 2025.
- [16] A. Sakin and et al. Bdsl-spoter: A transformer-based framework for bengali sign language recognition with cultural adaptation. *arXiv preprint arXiv:2501.00123*, 2025.
- [17] Kayo Yin and Jesse Read. Better sign language translation with explicit linguistic structures. *arXiv preprint arXiv:2004.00145*, 2020.
- [18] J. Kim and D. Lee. Augmenting sign language translation datasets with large language models. *Proceedings of the CVPR*, 2025.
- [19] Heiga Zen, Andrew Senior, and Mike Schuster. Statistical parametric speech synthesis using deep neural networks. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 7962–7966, 2013.
- [20] Xu Tan, Tao Qin, Frank Soong, and Tie-Yan Liu. A survey on neural speech synthesis. *arXiv preprint arXiv:2106.15561*, 2021.
- [21] M. R. I. Showrav and et al. Sign-to-speech: Generating natural language audio from skeletal sign language input using transformer models. *Multimedia Tools and Applications*, 2025.
- [22] T. R. Osmani and M. Ahmed. Two dimensional convolutional neural network approach for real-time bangla sign language characters recognition and translation. *International Journal of Computer Vision*, 2025.
- [23] BAUST Research Group. Baust lipi: A bdsl dataset with deep learning based bangla sign language recognition. *Journal of Computing and Information Technology*, 2024.
- [24] S. Ahmed and et al. Btvsl: A novel sentence-level annotated dataset for bangla sign language translation. *IEEE Access*, 2024.
- [25] Md. Al-Amin Sajib and Md. Saiful Islam. A deep learning approach for bengali text-to-speech synthesis. In *International Conference on Computer and Information Technology (ICCIT)*, pages 1–6, 2022.
- [26] Md. Saiful Islam and M. M. Rahman. Challenges in bengali sign language recognition and sentence construction. *Journal of Computer Science and Technology Studies*, 4(2):45–56, 2022.

References

References

- [27] P. Baruah and R. Dutta. Real-time assamese sign language recognition using mediapipe and deep learning. *International Conference on Emerging Technologies*, 2023.
- [28] Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, and Richard Bowden. Neural sign language translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7784–7793, 2018.
- [29] N. C. Camgoz and et al. Multi-channel transformers for multi-articulatory sign language translation. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020.
- [30] H. Wang and L. Zhang. Signbind-llm: Multi-stage modality fusion for sign language translation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
- [31] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 779–788, 2016.
- [32] Camillo Lugaresi, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Somu, Chuo Zhang, Amy Castro, Michael Russell, Juhyun Gopal, Yuan-Ting Chang, Tingbo Wang, David Zou, Oliver Chuo, Fan Zhang, Sean Tseng, Ming-Chang Sun, Jun Kwak, Caroline Wu, David Feng, Charles Tseng, and Matthias Grundmann. Mediapipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.
- [33] Chien-Yao Wang, Alexey Brosh Bochkovskiy, and Hong-Yuan Mark Liao. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7464–7475, 2023.
- [34] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 5998–6008, 2017.
- [35] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4171–4186, 2019.
- [36] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.

References

References

- [37] Gemma Team, Thomas Mesnard, Rustam Hardin, Robert Viola, and et al. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.
- [38] Zhi Zhou, Xu Chen, En Li, Liekang Zeng, Ke Luo, and Junshan Zhang. Edge intelligence: Paving the last mile of artificial intelligence with edge computing. *Proceedings of the IEEE*, 107(8):1738–1762, 2019.
- [39] Bjarne Stroustrup. The c++ programming language. 2013.
- [40] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 8024–8035, 2019.
- [41] Fan Zhang, Valentin Bazavan, Han-Pang Tsai, Yuan-Ting Chang, Sagar Shrestha, Juhyun Gopal, and Matthias Grundmann. Mediapipe hands: On-device real-time hand tracking. In *CVPR Workshop on Computer Vision for Augmented and Virtual Reality*, 2020.
- [42] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandeep Sandholm, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Benjamin Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1877–1901, 2020.
- [43] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. *Advances in Neural Information Processing Systems (NeurIPS)*, 36, 2023.
- [44] Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. Gptq: Accurate post-training quantization for generative pre-trained transformers. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2023.
- [45] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Xu, Yifan Yu, Zhipeng Chen, Yushuo Chen, Zifan Jiang, Ruiyang He, Yinfeng Wang, Jiapeng Ding, Jingyuan Wang, Jiacheng Liu, Zhejie Gu, Jiwen He, Zhouliang Liu, Zheng Huang, Zheng Zhao, Jiakai Zhang, Wanghe Huang, Ruhong Wang, Xiao Zhou, Xiaoxuan Chen, Jie Wang,

References

References

Jian-Yun Han, Xuanwei Hu, Yisong Ding, Pianpian Cao, Tingyu Wang, Zhenwei Shi, Junhan Zhang, Jixin Li, Jinshuai Kang, Zheng Zhang, Shangshuai Han, Xingyu Deng, Rui Wang, Qingxi Yuan, Jiakai Fang, Lidong Chu, Zhao Zou, Minting Zhang, Fengyan Zhu, Bowen Zhang, Hangyu Xu, Xiaoneng Yang, Duanyu Li, Runqi Wang, Jinhao Jin, Tongyu Yang, Tingting Zhou, Yuhao Shi, Shunkang Zhao, Wanbing Han, Kui Ding, Bowen Jin, Zelong Lin, Zhen Chen, Fan Yin, Dinghao Cao, Yanyan Liang, Yuanqi Wang, Ruirui Yao, Yihong Wang, Zhenyi Jin, Tong Wang, Zhen Jin, Yunjie Zhang, Kai He, Zeru Ding, Zhiyong Wang, Kai Ding, Jiarong Wang, Rui Jin, Xuan Wang, Jing Jin, Siyu Zhang, and Zemin Wang. A survey of large language models. *arXiv preprint arXiv:2303.18223*, 2023.